Memo to file

Subject: Survey of various projects in Kingston

Oct 28, '60

Met with R. Cypser 9:45 am.

Schedule:		
Low Temp:	Jim Beasley	10:00
Thin Film:	Carroll	
Compass:	Cunahan	
Informer:	Montagne	"
Map Compilation	Cliff Wilkins	2:00
Displays	Bottomley	2:45
R-32	pick up	

3 hrs to drive to KY,

Super Stretch Plan: Dr Cypser 9:45 am

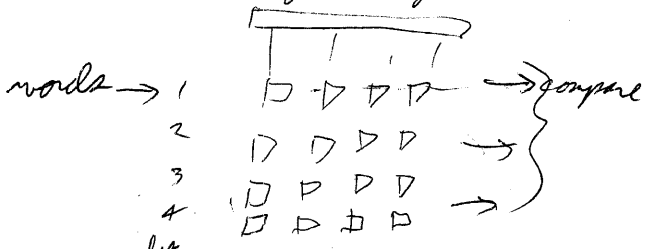
5 people initially to work with Research at Mahonai on Super Stretch.

At end of 2 years - Kingston would have more people than Research -- Takeover to make final assembly, -- all of Kingston has been reassessed -- FSD to displace DSD -- even in manufacturing?

Machine Organization: Behnke

group works on various machine technologies but leaning toward cryogenics: they hope to get 50¢ per bit eventually,

Associative Memory: (Tag Memories)



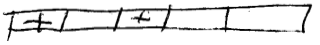
Associative Memories 10 to 20 times as long cycle in core memory. (have to look at bit by bit)
Cryogenics are slow but not that much slower

Cryogenics can give:

"greater than"	≥ Iw
"less than"	≤ Iw
"range"	interval (using both)

Associative memory
can read destructively or non-destructively,
selective write any bit position by masking

Logical and/or Function



ie, compare on certain fields only.

Length of fields?
- can be any length -- the longer the slower in cryogenics

Rex Seeber has used assoc. mem. for sorting routines
2.7 N memory cycles for sorting. (obtained by simulation on 704)

Other Logical functions:
"next highest"
"next lowest"
"nearest"
} are working on this — most challenging problem not yet worked out

Proposal to Rome: 4 words - 2 bits ea, cryogenic associative memory. There is a good chance of getting contract.

McDermott at Nebraska -- core mem. assoc. mem. of small size has been tried?
- can't do greater than ~ten words etc - need Transistors, at each bit position for logic

Asked about these possibilities for them to consider

Question of "side-ways" mem. →  read-write either way...

" " CRT mem. } — do correlation — 2 deflector voltages
— find nearest values

more natural relation — high density-low cost
less reliability regs.

Should an effort go into this?

Jim Beasley - cryogenics

200 mil

are much faster ---

Ja-Lins cryotrons - tested last June . 2 ns 30 Mc repeate

→ 40 bit mem plane . 135 cryotron (see This.)

(57⅓ per sq inch)

used 17 masks, 19 evaporations

mostly etched glass masks,

Drawings 1. initial drawing. large on graph paper

2. 10x rule — artwork, accurate

3. cut "rubylith" master 10x -- more accurate than drawing

4. "step & repeat" camera (1 mil in 2 feet area)

5. film reproduction to glass plate

6. Take to Corning etched glass form - (two months total for all masks on last model)

above takes 4 months total Time

most is Time sequence — hard to parallel

Carroll -- Thin Film activities.

contract Sig C Jul'59 - Jul'60

multilayered thin films - passive elements capacitors resistors etc

3 emitting type not memory devices.

3 appl types. not much luck with inductors

layout — less feed back needed ---

- micro transistors — could be incorporated with these type units.

30 or 40 circuits - mass soldering techniques.

— Temperature 200°F 80 mwatts dissipation ---

mixtures; not as great but has better temperature characteristics
- good contact with surface
- also materials themselves are stable with Temp, - Aluminum

- multivibrator circuits --- made 160 of them for yield study
made 40 per week 138 of them worked
86%

Resistors: ±15%
Capacitors: ±25%

500 KC operation free-running multivibrator,

⇒ a final report now being written on above prog.
{ Vacuum deposition of three materials
SiO_2, Nichrome, Al.

Environmental Tests --- Temp variations, humidity
Thermal shock test -- some trouble after 13 cycles at high humidity. peeling of film

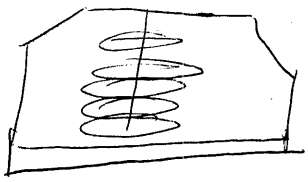
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Plan to manufacture large no. of circuits
Transmission line Type connections
prototype model in 1 year or 2

Very high speeds? -- can deposit transmission lines
- can use high speed-small trs.
100 ns 10 Mc rep rate -- rise times must faster -- depends on Trs, resistors are stable with freq,

Company Counihan, Glenn

See ⇒ Proposal # 9698 19 Sept 60



4096 6 volt supply

double aperture ferrite core "wampum belt"

Discs 3" dia.

solder fillet joints.

Schedule

Phase I { 12 mos – feasibility model – not complete (short word, small mem)
24 mos – prototype – flyable, full scale

Phase II 18 mos 6 machines – one every 2 mos during

Phase III 12 mos 12 machines – one per mo

hope to go to direct deposition

late deposit active elements – several years off – maybe 10?

[3rd quarterly Report Contract DA 36-039-sc-84547
30 April 60

Need additional Development -- (interlocking coverage)
e.g. – a small system -- weight or space reg. -- for special case } ← several million needed

– see paper on max. density interconnections

Map Program - for C of Engrs (Ft. Belvoir)
- high speed demonstration on 704 will be done. (Oct-Nov 61)

stereo-pairs - aerial photos
- originally had analog scanner
- then used 500 spots/inch - densitometer - recorded on tape
160 shades of gray could be determined (eye gets up to 100)
pattern match.
(Hooker-Sancto, worked out)
calculate altitude
compute orthographic map, - also altitude of each spot - write on tape,
print contours etc
print off-line analog plotter

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other capabilities - Stereo-X-Rays with Tulane Univ.
also more shades of gray

Field Data: Stem-Mattingly
"hard" computer - mobile van mounted
20-40 K instrs per sec
38 bit (1 parity, 1 sign) (Moby-Dick op code)
26 μs add.
200 μs mpy

has disk file
- data retrieval - (present contract) { Control, Logistics, Intelligence
- comm. line switching

small size - no air cond.
- 25°F, + 132°F

heat problem - mag tape core logic - pulses - low power & rugged.
(+12° above ambient -- gets close to Tc, germanium danger points.

Unit Testing - starts in week or so.

- memory built (30-50 Telemeta magnetics)
- memory is kept heated ~~@~~ to 40°C, 2° max rise due to programs.
- 8 μsec memory

Tac on Jan 1

I/O comerter Feb 1

Multi channel comm. comerter -- scanning several devices, - ~~@~~ being planned,

= higher speeds are not practical -- 1,5 Mc clock about max.

Fodac - a very small

Minipac - will be compatible

Moby Dick - top speed (7 or order)

Philco ~~Basic~~ Pac - limited op codes

Displays - Bottomley

- no hardware now --

committee few mos ago -- all corporate Divs

Presentation to RTD Board. -- what are reqs? from each division

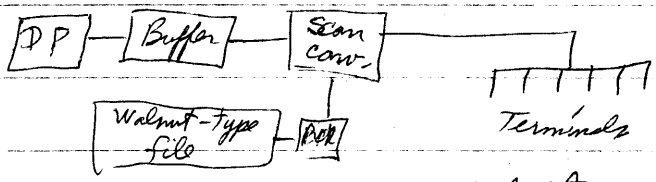
FSD proposed -- for commercial

- corporate responsibility. (DSD ?) Market and, Systems Spec.
- Display Technology DSD initially → DSD-Computer
- Engineering Area, FSD

Production

Synergy plan:

(1) Direct view CRT < GEM Mod Multi-Terminal System



Mylar Storage Tubes ← ASDD Many Tech?

~$1000 cost? TV raster scan

above all to start 1961. (Mostly MARP)

Q-32 Type Display: - command-control -- still has good hope

(2) Projection system optical printer projector < alpha-Numerical symbols
over-lay capability

Kalfax - infra Red fixing
½ sec to fix
(40 char/sec writing rate)

CRT → Selenium → Kalfax - high writing rates
- two step process. ½ sec to transport